

One Model, Many Geometries: Spectral Dynamics of Multilingual LLMs

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Abstract

We explore the geometric evolution of multilingual LLM representations during pretraining, tracking per-language RankMe across 50+ checkpoints for two multilingual models and 10–15 diverse languages. We ask whether geometric phases align across languages, what drives any differences, and whether divergence predicts downstream performance.

1 Introduction

While loss decreases monotonically during LLM pretraining, Li et al. (2025) show that representation geometry evolves in three non-monotonic phases. However, their analysis is limited to English models, and multilingual training dynamics have largely been studied using only feature-level methods or sparse checkpoints (Blevins et al., 2022; Wang et al., 2024; Chang et al., 2022).

We hypothesize that the macro-level evolution of per-language representations predicts multilingual capabilities. To test this, we track per-language spectral measures across 50+ checkpoints for two multilingual models, asking whether (1) languages share geometric phases, (2) differences are explained by language properties or data distribution, and (3) these differences predict downstream performance and cross-lingual transfer.

2 Approach

Models. We use two open-source 8B multilingual LLMs: Apertus-8B (Apertus et al., 2025) (71 checkpoints), and FuxiTranyu-8B (Sun et al., 2024) (58 checkpoints).

RankMe. Our primary metric for analyzing geometry evolution is RankMe (Garrido et al., 2023), a measure of effective rank also used by Li et al. (2025), whose definition we adopt. For a layer L with hidden dimension D , we collect $N \gg D$ activations (e.g., the last token from N sequences)

into a matrix X . We then compute the centered covariance matrix Σ , and normalize its eigenvalues λ to yield a probability distribution $p_k = \lambda_k / \sum_j \lambda_j$. RankMe is the exponentiated entropy of p : $\text{RankMe}(X) = \exp(-\sum_k p_k \ln p_k)$. We compute RankMe for a fixed L across all checkpoints of both models for 10–15 diverse languages.

Datasets and benchmarks. We compute RankMe on language-specific subsets of FineWeb2 (Penedo et al., 2025). We use XCOPA (Ponti et al., 2020; Roemmele et al., 2011) and m-MMLU (Hendrycks et al., 2020) to evaluate downstream multilingual abilities, and ECLeKTic (Goldman et al., 2025) to evaluate cross-lingual transfer. We also test zero-shot transfer using pairs of language-specific subsets from XNLI (Conneau et al., 2018).

3 Project Timeline

In **Phase 1** (May 3–May 24), we perform representation extraction and geometry analysis, implementing the pipeline to extract RankMe metrics from our checkpoints and mapping the geometric evolution of all selected languages. In **Phase 2** (May 25–June 7), we evaluate capabilities and synthesize results, correlating geometric phases with language properties and downstream monolingual and cross-lingual performance. Each of us will explore different properties that RankMe curves may be correlated with:

- **João:** Do multilingual LLMs exhibit synchronized geometric phases across languages during pretraining?
- **Pedro:** Do RankMe curves depend on language families or training data distribution?
- **Giacomo:** Do properties of RankMe evolution predict downstream monolingual performance?
- **Mattia:** Does the difference between curves predict cross-lingual transfer?

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